

9708/33/o/n/23

1-D:

When consumption increases from **1 unit to 4 units**, **Marginal utility is diminishing but is positive**, which represents that **total utility will rise but at a decreasing rate**.

When consumption increases from **4 units to 5 units**, **Marginal utility is zero**, which represents that **total utility is at its maximum**.

When consumption increases from **5 units to 6 units**, **Marginal utility will be negative**, which represents that **total utility will fall**.

That being said, as consumption increases from 1 unit to 6 units, total utility will rise and then fall.

2-B:

1. **Diminishing marginal utility:** As a consumer consumes more of a good, the additional utility (satisfaction) gained from each additional unit decreases. This means that the marginal utility of each additional unit falls. **This gives existence to decreasing opportunity cost.** This is because, as the marginal utility of a good falls, the opportunity cost of choosing that good over another good also falls. This means that the consumer is willing to give up fewer units of the other good to get an additional unit of the first good.
2. This leads to a **diminishing marginal rate of substitution**. This is because, the rate at which, a consumer is willing to give up a good in exchange for the other decreases. (**Note that MRS is the slope of the indifference curve.** It's the rate at which a consumer is willing to trade one good for another, while maintaining the same level of satisfaction or utility.)

This explains the reason why indifference curve is convex to the origin.

A. Allocative efficiency: Refers to the optimal allocation of resources in an economy that occurs when $MSB=MSC$.

C. Pareto optimality: A situation where no individual can be made better off without making someone else worse off.

D. Equi-marginal principle: Represents the utility maximizing behavior of a rational consumer that occurs when a consumption bundle that maximizes total utility is consumed. This is when:- $Mux/Px = MUY/ PY$.

3-C:

The principal-agent problem refers to the challenge of aligning the interests of agents (managers or directors) with those of the principal (shareholders or owners). When managers are not allowed to hold shares, they may not have a direct stake in the firm's profitability, which can exacerbate the principal-agent problem. **This explains why key C will be correct since it would further create a conflict of interests between principal and the agents.**

A. Ensuring the actions of agents are known by the managers (transparency) would actually help reduce the principal-agent problem.

B. Linking the rewards of managers with the profitability of the firm (performance-based compensation) would align the interests of agents with those of the principal, reducing the principal-agent problem.

D. Preparing contracts that compel agents to work in the interest of the principal (contractual alignment) would also reduce the principal-agent problem.

4-A:

Output (Q)	Total Fixed Cost (TFC)	Total Variable Cost (TVC)	Average Fixed Cost (AFC)	Average Variable Cost (AVC)
1	\$1000	\$1000	$AFC = \$1000 / 1 = \1000	$AVC = \$1000 / 1 = \1000
2	\$1000	\$2000	$AFC = \$1000 / 2 = \500	$AVC = \$2000 / 2 = \1000
3	\$1000	\$3000	$AFC = \$1000 / 3 = \333.33	$AVC = \$3000 / 3 = \1000
4	\$1000	\$4000	$AFC = \$1000 / 4 = \250	$AVC = \$4000 / 4 = \1000
5	\$1000	\$5000	$AFC = \$1000 / 5 = \200	$AVC = \$5000 / 5 = \1000
6	\$1000	\$6000	$AFC = \$1000 / 6 = \166.67	$AVC = \$6000 / 6 = \1000

Formulae used:

1. $AFC = TFC / Q$

2. $AVC = TVC / Q$

As the output level increases, the average fixed cost (AFC) decreases, since the total fixed cost is spread over a larger quantity of output. This explains that only key A can be correct.

5-A:

➔ Since an improvement takes place in manufacturing techniques, there will be a decrease in average costs achieved by the firm over time, because of which, the average costs at the current level of output will fall. This explains that there will be dynamic efficiency.

To reflect this, there must be a downward shift in the average cost curve from ATC1 to ATC2.

➔ Additionally, because of an improvement in manufacturing techniques in industry X, the costs of production of firms will fall, which will bring about an increase in supply in the market for good X.

This can be shown through a rightward shift in the supply curve from S3 to S4.

As a result of which, there will be a surplus in the market for good X, which will bring about a fall in prices, consequently resulting in an extension in quantity demanded (equilibrium quantity)

→ Technological improvements in industry X, since improve the quality of capital (which is a factor of production), will result in a rightward shift in the PPC. However, the shift will be such that there will be an increase in output of good X.

This can be seen through the PPC shifting from P5 to P7. In other words, the new PPC will be P7.

6-C:

The correct answer is C: vertical backward integration.

Vertical backward integration occurs when a company acquires a supplier that provides inputs for its products. **In this case, the coffee restaurant chain acquired a coffee bean farm, which is a supplier of raw materials (coffee beans) for their business.** This allows them to control the supply of high-quality coffee beans and reduce dependence on external suppliers.

A. Horizontal integration: This occurs when a company acquires a competitor or a company at the same stage of production.

B. Organic growth: This refers to growth achieved through internal expansion, such as opening new locations or increasing production capacity, without acquiring other companies.

D. Vertical forward integration: This occurs when a company acquires a distributor or a company that is involved in the next stage of the production process.

7-B:

Third-degree price discrimination occurs when a firm charges different prices to different consumer groups for the same product, based on differences in price elasticities of demand. To practice third-degree price discrimination effectively, a firm must be able to segment its market into distinct groups with different price elasticities of demand. This means that the firm must be able to identify and separate different customer groups, such as students, seniors, and adults. **This explains why key B will be correct.**

A. This is not a necessary condition for third-degree price discrimination. Firms can segment their market based on other characteristics, such as age, income, or occupation, regardless of geographical location.

C. The firm relies on differences in price elasticity of demand across segments to charge different prices.

D. Price undercutting is required to enter a monopolistic market: This is not related to third-degree price discrimination, which is a pricing strategy used by firms with market power, not a strategy for entering a market.

8-B:

Important note:- By switching to motorcycles, the individuals who make the change experience a private benefit (faster journeys), while also generating external benefits for others (faster journeys for car drivers, increased safety for cyclists, and reduced air pollution for pedestrians).

A private benefit is a benefit that accrues directly to the individual who makes a change, in this case, switching to a motorcycle. The benefit of faster journeys for those who switch to motorcycles is a direct benefit that they experience themselves. **This explains why this will be a private benefit, therefore leading to option B being the correct option.**

A. Faster journeys for those who continued to go by car: This is an external benefit, also known as a positive externality. **This is because, people who continue to use the car benefit from the action of the car drivers who have switched to motorcycles.** This explains that this will be a third party benefit.

C. This is also an external benefit, as it benefits cyclists who are not the ones switching to motorcycles.

D. Reduced levels of air pollution for pedestrians: This is another external benefit, as it benefits pedestrians who are not directly involved in the change.

9-A:

Government Cost-benefit analysis is a decision-making tool that compares the SB of a project or decision to SC. The project which yields the highest net social benefit is selected. However, its accuracy can be reduced by certain conditions:

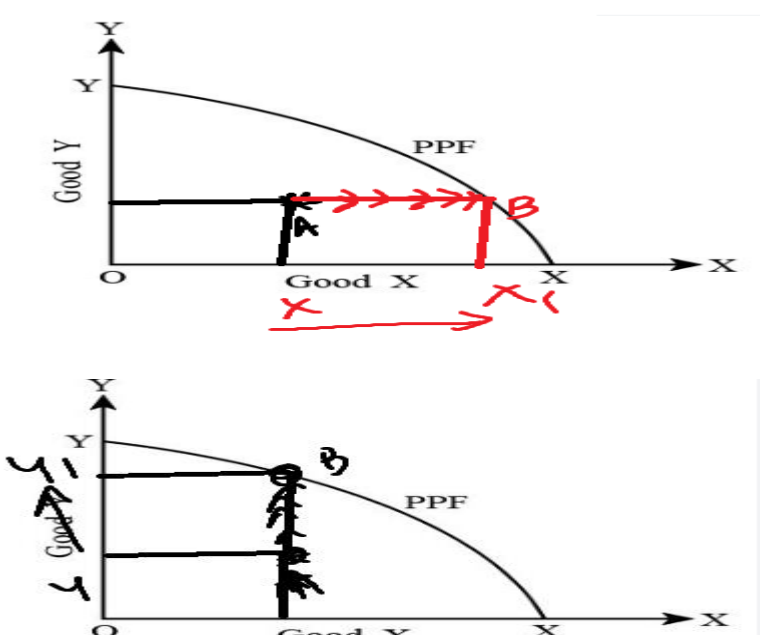
- High proportion of external costs and benefits to total costs and benefits (i.e., high externalities): Externalities are costs or benefits that affect third parties, not directly involved in the decision. When externalities are high, it can be difficult to accurately quantify and account for them. **This is because, externalities do not go through the market, which is why, it becomes difficult to attach a price to them.**
- High percentage of total costs and benefits that are in the future (i.e., high time discounting): Costs and benefits that occur far in the future are typically discounted to their present value. However, the further into the future, the more uncertainty and risk are involved, making it harder to accurately estimate costs and benefits.

10-C:

Pareto optimality is a state where resources are allocated in such a way that no individual can be made better off without making someone else worse off. In other words, it's a situation where the allocation of resources is efficient, and any changes would lead to a loss for at least one individual.

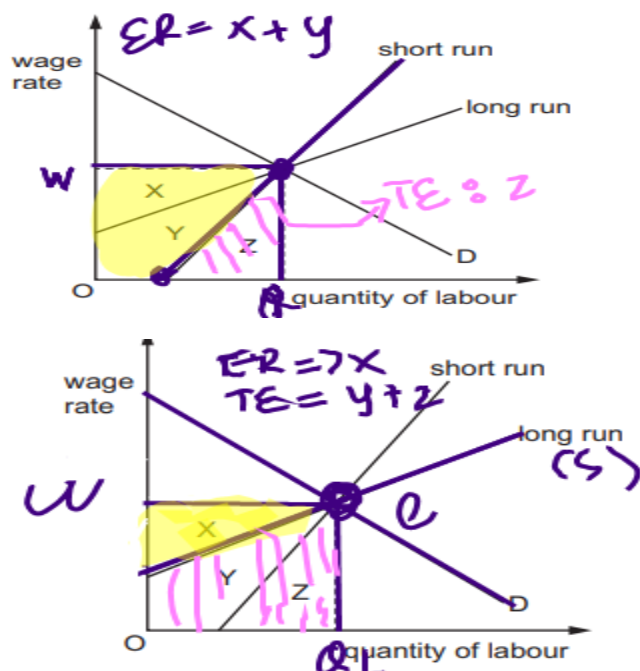
Since the question says that there is a movement towards Pareto optimality, this implies that currently we are in operating inside the PPC, and there is a move from the Pareto inefficient point to the Pareto efficient point, that is on the PPC.

Therefore, key C will be correct because when we are moving towards the Pareto optimal point, this means that **re-allocation of resources will can increase the output of one good without decreasing the output of another good.**



From an individual's perspective, a move towards Pareto optimality would represent that reallocation of resources can make someone better off without making any other person worse off.

11-C:



Transfer Earnings:

Transfer earnings are the minimum amount that a factor of production (e.g., labor) must receive to remain in its current occupation or industry. It's the opportunity cost of transferring to another occupation or industry. **It is seen through the area under the supply curve.**

Economic Rent:

Economic rent, on the other hand, is the excess earnings that a factor of production receives above its transfer earnings. It's the surplus value that a factor of production generates due to its unique or scarce qualities. **The economic rent is represented by the area above the supply curve and below the equilibrium price (wage)**

It can be seen in the diagram above that region Y that was previously economic rent in the short-run has become transfer earnings in the long-run.

12-D:

The **marginal productivity of labor (MPL)** is the change in total output that results from a one-unit increase in labor input, holding all other inputs constant. In other words, it measures the additional output generated by adding one more unit of labor to the production process.

MPL is calculated as:

$$MPL = \Delta TP / \Delta L$$

- To increase the **marginal productivity of labor**, a government might introduce policies such as:-

- Better education: Yes, investing in education and training programs can enhance workers' skills and knowledge, leading to higher productivity.
- Tax allowances for capital goods: Yes, providing incentives for businesses to invest in new capital goods, such as machinery and technology, can improve worker productivity.

13-D:

D) Removing the national minimum wage would likely increase relative poverty in a country because it would allow employers to pay lower wages to unskilled lowly-paid workers, which could lead to a decrease in the income of low-wage workers relative to the high wage ones who remain unaffected by this policy. **This explains how relative poverty is likely to increase.**

A. Increasing means-tested benefit levels: This would likely reduce poverty, as it would provide more financial support to those in need.

B. Increasing the power of trade unions: Trade unions advocate for better wages, and if they end up persuading the government to pass a minimum wage legislation, the wages earned by unskilled workers is likely to increase, which will reduce the wage gap between skilled and unskilled workers. Therefore, relative poverty will fall.

C. Making the tax system more progressive: A more progressive tax system would require the wealthy to pay a higher tax rate, reducing income inequality and potentially reducing poverty.

All the other options are incorrect because they reduce relative poverty by reducing rich-poor disparities.

14-D:

Import quotas are a trade policy tool used to limit the quantity of goods imported into a country. As being a protectionist tool, it may be used to protect domestic industries, but it is not typically designed to correct a market failure. **This explains why key D will be correct.**

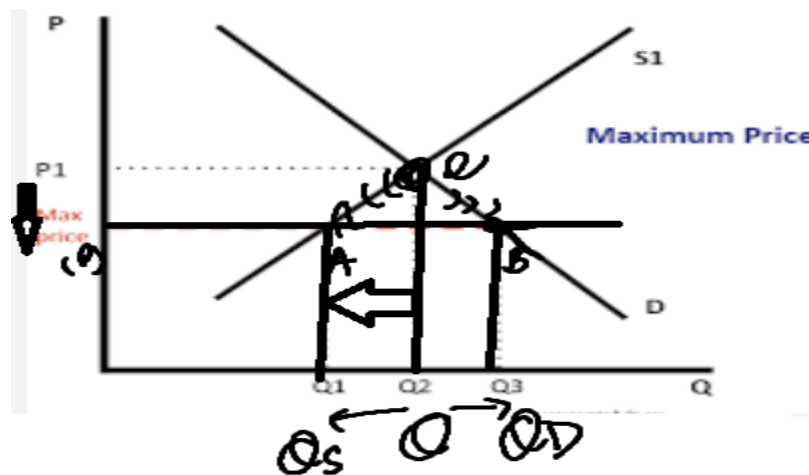
A. Government provision of healthcare: This corrects the market failure associated with under-allocation in the market of merit goods.

B. Price controls on large firms: This corrects the market failure of monopoly power (large firms may exploit their market dominance to set high prices).

C. Regulations to limit smoke pollution: This corrects the market failure associated with over-allocation of demerit goods with negative externalities (pollution affects third parties, not just the firm producing it).

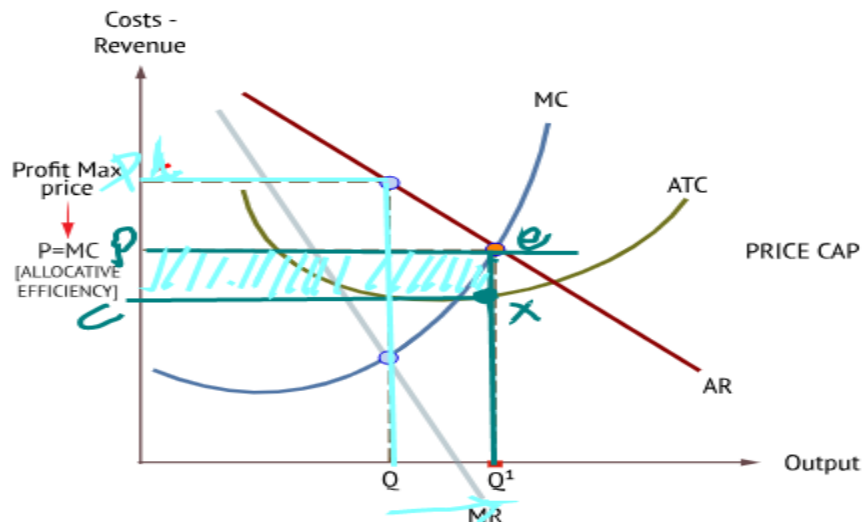
15-D:

When a government imposes a maximum price (price ceiling) below the free market price, it can lead to a shortage of the good. This is because suppliers are not willing to supply the good at a price that is below the equilibrium price, leading to a reduction in the quantity supplied. On the other hand, at a low price level, quantity demanded extends. **This translates into shortages, as a result of which, key D becomes correct.**



A. Maximum prices are typically associated with a decrease in inflation as price rises are limited.

B. Price ceilings actually reduce profits, as firms are forced to sell at a lower price.



When because of a price-cap, a profit maximizing monopoly is forced to reduce prices to where $MC=AR$, profits are reduced. This can be seen above.

C. Price ceilings lead to shortages, not surpluses (refer to explanation of part d above)

16-C:

An increase in national income is most likely to induce an increase in investment spending when firms believe that the increase in income is permanent, indicating a long-term growth trend. This confidence in sustained growth encourages firms to **invest in new capital in order to meet future demand**. This explains why key C is correct.

A. Spare production capacity: Having spare capacity might actually lead to reduced investment, as firms may not need to expand capacity to meet demand.

B. Labor-intensive production: If production is labor intensive, firms may actually not have to increase their investment in capital goods.

D. Temporary increase in income: If the increase is seen as temporary, firms may not invest in new capacity or projects, as the demand surge is expected to be short-lived.

17-C:

The functions of money are: **medium of exchange, store of value, unit of account, and standard of deferred payments**.

A. This option is incorrect because while money can be divided into smaller units (e.g., dollars into cents), **divisibility** is a characteristic of money, not a function.

B. This option is also incorrect because, like divisibility, **portability** is a characteristic of money, not a function. Cash is often lightweight and easy to carry, but this convenience doesn't constitute a function of money.

C. This option is correct because money serves as a **standard of deferred payment**, allowing for credit and future settlements. This means that money can be used to pay for goods and services now and settle the debt later, facilitating transactions and economic activity. This explains why key C is correct.

D. **General acceptability** is a characteristic that enables money to perform its functions, but it's not a function like medium of exchange or store of value.

18-C:

The shift to remote work leads to a decline in demand for services and businesses in city centers, potentially resulting in:-

- Permanent closures of coffee shops, retailers, and other businesses -> Job losses in industries that rely on in-person interactions
- **To sum up, there will be** a structural change in the economy, as some jobs become obsolete and are replaced by new ones.
- **Note: structural change occurs when there is mainly a shift in the pattern of demand and supply in the economy..**

This scenario is likely to lead to structural unemployment, as workers in affected industries may need to acquire new skills to adapt to the changing job market.

This explains why key C will be correct.

A. Frictional unemployment occurs when workers are unemployed temporarily while they are in between jobs. Remote work doesn't directly affect frictional unemployment.

B. Remote work reduces the importance of geographical mobility of labor.

D. Voluntary unemployment occurs when workers choose not to work at the current market wage rate. The shift to remote work is unlikely to lead to an increase in voluntary unemployment.

19-A:

A. Allowing more labor immigration can help China address its declining workforce and aging population. By bringing in new workers, China can increase its labor force size, and offset the impact of the declining workforce. This can help sustain economic growth.

B. Encouraging women to leave work to have more children might increase the birth rate, but it would likely reduce the labor force in the short term. This could exacerbate the issue in the short-term.

C. In the short term, labor immigration could be a more effective way to address the labor force decline, rather than an increased research in technology to improve labor productivity which might be a long-term solution.

D Taxing the labor force more to pay for pensions could have negative economic consequences, such as this would reduce disposable incomes and subsequently consumption and Aggregate demand will dampen, leading to a negative impact on economic growth.

20-D:

A reduction in defense expenditure means that the government is spending less, and this decrease in autonomous injection into the circular flow will lead to a decrease in aggregate demand, and subsequently, there will be a decrease in national income. **This explains why key D is correct.**

A. Savings are a leakage from the circular flow of income, so a fall in planned savings means that more money is being spent, leading to an increase in national income.

B: A decrease in imports means that more domestic production is being consumed, leading to an increase in Aggregate demand and subsequently national income will rise too.

C. An increase in the proportion of income that goes in consumption leads to an increase in aggregate demand, and subsequently, an increase in national income.

21-B:

The **accelerator principle** states that investment is driven by changes in national income. According to the accelerator principle, investment is a function of changes in national income.

Therefore, the co-efficient of the accelerator can be calculated as: Change in investment/change in national income.

When national income grows at a slower rate, businesses anticipate lower future demand, leading to a decrease in induced investment. **This explains why key B will be correct.**

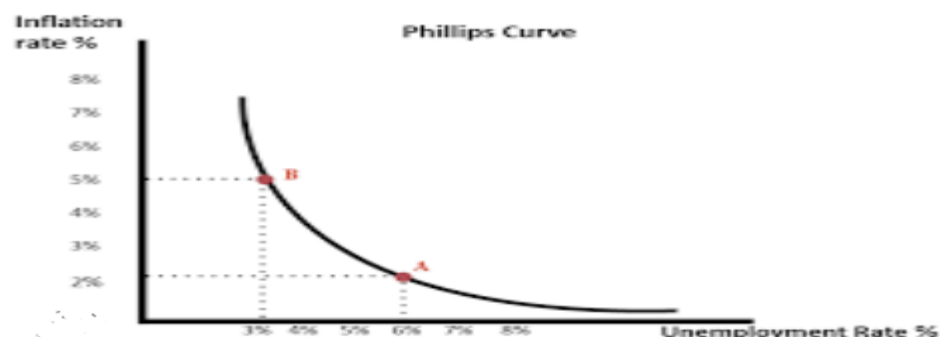
A. A decrease in confidence: While a decrease in confidence can lead to a decrease in investment, it's not the specific factor identified by the accelerator principle.

C. An increase in the price of capital equipment: An increase in price might affect the cost of investment, but it is not linked to the accelerator principle.

D. An increase in the rate of interest: An increase in interest rates can increase cost of borrowing and subsequently reduce investment, but it's not the specific factor identified by the accelerator principle. **When we observe the impact of a change in interest rate on investment decisions, the concept of marginal efficiency of investment comes in.**

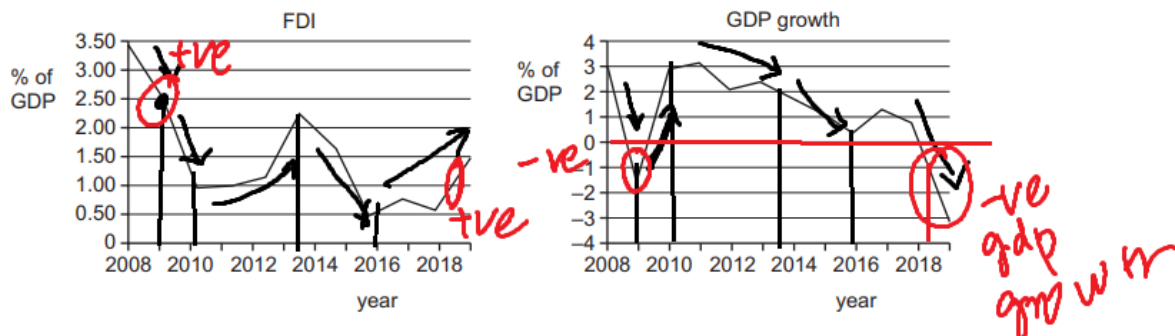
22-C:

The Phillips curve is a **historical inverse relationship** between the rate of unemployment and the rate of inflation. The vertical axis measures the rate of inflation, while the horizontal axis measures the rate of unemployment. **This explains why key C is correct.**



23-A:

South Africa has attempted to improve the level of economic growth by attracting foreign direct investment (FDI).



No proper relationship can be identified between FDI and the level of economic growth.

2008-2009: Both, foreign direct investment as a percentage of gdp and GDP growth as a % of gdp fell. This represents a positive relationship between both of them.

2009-2010: Foreign direct investment as a percentage of gdp fell whereas GDP growth as a % of gdp rose. This represents a negative relationship between both of them.

This subsequently proves that no proper trend is identifiable, hence making key A correct.

- B) Increases in FDI are not always linked with improvement in GDP growth. This is because, it can be seen in explanation of part A that there is no proper relationship between both of them.
- C) Lowering FDI is not linked to stable gdp growth. This is because, it can be seen that when FDI as a % of GDP fell in between 2008-2009, the gdp growth rate fell too.
- D) It can be seen when there was negative gdp growth (in 2009 and after 2018), there were positive inflows of Fdi as a % of gdp. This explains why key D will be incorrect.

24-B:

Economic development is a broad concept that encompasses not only economic growth but also improvements in living standards, health, education, and overall well-being. An improvement in the average citizen's quality of life is a strong indicator of economic development.

A. Appreciation of the exchange rate will lead to a Balance of trade deficit in the long-run, which will result in a fall in Aggregate demand, as a result of which, there will be negative economic growth. This will also be detrimental for economic development.

C. A trade surplus will result in net injections into the circular flow, because of which, there will be an increase in Aggregate demand and consequently actual economic growth, but it doesn't directly measure improvements in citizens' quality of life.

D. GDP growth is important, but it doesn't necessarily translate to improvements in living standards or quality of life.

The difference between economic growth and economic development is outlined in the table below:

Definition	Increase in a country's GDP or output	Improvement in living standards, well-being, and quality of life
Focus	Quantitative expansion of economy	Qualitative transformation of economy and society
Indicators	GDP, GNP, Per Capita Income	Human Development Index (HDI), Poverty Rate, Life Expectancy, Education (e.g., literacy rates, enrollment), Healthcare (e.g., infant mortality, access to healthcare), Income Inequality (e.g., Gini coefficient), Access to Basic Services (e.g., water, sanitation, electricity), Gender Equality, Environmental Sustainability (e.g., carbon footprint, renewable energy)
Scope	Narrow, focuses on economic output	Broad, encompasses economic, social, and political aspects

25-D:

The table gives some economic indicators for four countries.

Which country is closest to achieving the principal aims of government economic policy?

	* change in GDP (%)	* rate of inflation (%)	* unemployment (%)	* balance of payments (% of GDP)
A	0.47	0.8 ^{2nd}	3.4 ^{1st}	+3.3 ^{2nd}
B	1.13	0.9	10.5	-0.1
C	1.18 ^{2nd}	1.1	6.9	-3.3
<u>D</u>	<u>1.45</u> ^{1st}	<u>0.1</u> ^{1st}	<u>4.6</u> ^{2nd}	<u>+8.5</u> ^{1st}

~~*~~ → The higher the better
 * → The lower the better

The key macro-economic objectives are: re-distribution of income and wealth, economic growth (measured through changes in gdp), price stability (measured through rate of inflation), full employment (measured through unemployment%), balance of payments(measured through % of GDP)

Country D is closest to achieving all the four objectives because:

- Change in GDP is the highest
- Rate of inflation is the lowest.
- Unemployment% is the second lowest
- Balance of payments as a % of gdp is the highest.

26-A:

A fall in the general price level, also known as price stability is a common macroeconomic aim for governments. This means that the government aims to keep the rate of inflation low, usually around 2-3% annually, to maintain the purchasing power of consumers' money and promote economic stability. **This explains why key A is correct.**

B. While governments may aim to regulate monopolies to promote competition, this is a micro-economic intervention to solve market failure associated with monopolistic exploitation.

C. Lower taxes on consumer goods: This is a microeconomic concern, as it affects the prices and demand for specific goods and services ("**consumer goods**")

D. A minimum wage for low-paid agricultural workers: This is a microeconomic concern, as it focuses on the labor market and wages for a specific industry (**agriculture**).

27-B:

Economy Type	Dependency	Gini Coefficient	Per Capita Real Income
Low-Income	Primary sector (agriculture, mining)	High (close to 1)	Low
Middle-Income	Secondary sector (manufacturing, construction)	Medium (around 0.4-0.6)	Medium
High-Income	Tertiary sector (services, finance)	Low (close to 0)	High

- Dependency on the primary sector (agriculture, mining, etc.): Low-income economies often rely heavily on primary commodities for exports.
- High Gini coefficient: Low-income economies often have significant income inequality, indicating a high Gini coefficient (close to 1).
- Low per capita real income: By definition, low-income economies have low average incomes, indicating a low per capita real income.

This explains why key B is correct.

28-C:

The International Monetary Fund (IMF) plays a crucial role in:

- Providing short-term loans to countries facing balance of payments difficulties or foreign exchange shortages.
- Helping countries address economic crises, such as currency fluctuations or debt issues.
- Offering technical assistance and policy advice to member countries.

This explains why key C is correct.

Here's why the other options are not correct:

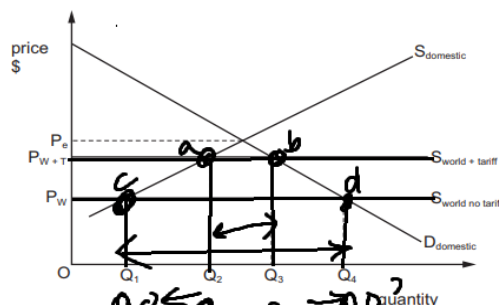
A. The World Trade Organization (WTO) is responsible for addressing trade disagreements, not the IMF.

B. The IMF is not directly involved in providing emergency assistance to countries in need, which is typically handled by organizations like the United Nations or non-governmental organizations (NGOs).

D. While the IMF does provide financing to developing countries, it is not interest-free, and the terms and conditions vary depending on the program or facility. **Additionally, mainly it is the World Bank who provides support to developing countries for infrastructure purposes.**

29-B:

The diagram shows the international trading position of a country that had tariffs on imports. The country removed the tariffs on imports.



What was the quantity of imports into this country after the removal of the tariffs?

- A Q_1Q_2 **B** Q_1Q_4 C Q_2Q_3 D Q_3Q_4

After tariff $\rightarrow$
 • price $\Rightarrow$ world price + tariff $\Rightarrow P_W + t$
 • $Q_S \Rightarrow Q_2$
 • $Q_D \Rightarrow Q_3$
 • volume of imports (shortage)
 $\Rightarrow Q_2 - Q_3$

After removal of tariff $\rightarrow$
 • price $\Rightarrow$ world price $\Rightarrow P_W$
 • $Q_S \Rightarrow Q_1$
 • $Q_D \Rightarrow Q_4$
 volume of imports (shortage)
 $\hookrightarrow Q_1 - Q_4$

30-C:

Option A: Central bank independence is important for monetary policy and economic stability. This means that it will not prevent FDI from coming in the manufacturing sector of the country.

Option B: Privatization can attract foreign investment, as it offers opportunities for companies to acquire assets and participate in the economy. Therefore, this option is not a reason why low wages and low-cost land may not attract foreign investment in manufacturing.

Option C: A lack of well-developed infrastructure, such as roads, ports, airports, reliable electricity, and water supply, can significantly increase production costs and reduce efficiency. This can make it difficult for companies to operate effectively, even with low wages and low-cost land. Therefore, this option is a valid reason why low wages and low-cost land may not attract foreign investment in manufacturing. **This explains why key C is correct.**

Option D: Permitting repatriation of profits is often a requirement for foreign investors, as they seek to repatriate their earnings back to their home country. If a country allows repatriation of profits, it can attract foreign investment.

